

ONLINE-ONLY SUPPLEMENT

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Online-Only Supplement 1 – eText 1

eText 1 Characteristics of randomized controlled trials with the potential to meaningfully inform practice.

1. Report of specific primary outcome(s)
2. Alignment of the trial design with the reported trial hypothesis for the primary outcome (e.g., superiority, non-inferiority or equivalence)
3. Definition of a clinically relevant effect as a target difference for the primary outcome(s) – i.e., a minimal important difference (MID) or a Cohen's d – to base sample size calculations upon
4. Consideration of clinical relevance (e.g., through the MID) to discuss and interpret the results for the primary outcome(s)

Online-Only Supplement 2 – eText 2

eText 2 Decision rules for the assessment of the four-domain composite endpoint.

Domain 1: Report of specific primary outcome

Labels	Decision rules
Yes	report of specific primary endpoint(s), in any resource (i.e., pre-trial documentation or final publication). If >1 resource, concordance is required.
No	- no report of specific primary endpoint(s) - report of multiple primary endpoints without multiplicity adjustment
Discordant between pretrial document and final RCT	quantitative and/or qualitative mismatch in the endpoint(s) reported

Domain 2: Alignment of trial design with reported trial hypothesis

Labels	Decision rules
Yes	description of trial design and/or statistical analyses aligned with the trial hypothesis reported for primary endpoint(s) in any resource (criterion 2.1 fulfilled). If >1 resource, concordance is required.
No	- mismatch between the hypothesis reported and the hypothesis corresponding to the description of trial design and/or statistical analyses - description of trial design and/or statistical analyses not aligned with a specific hypothesis for primary endpoint(s) in any resource - no description of trial design and/or statistical analyses for primary endpoint(s) in any resource
Discordant between pretrial document and final RCT	mismatch in the reported trial hypothesis and/or the trial hypothesis corresponding to the trial design and/or statistical analyses
Not applicable	- no report of specific primary endpoint(s) (condition 1 not fulfilled) - no report of specific hypothesis for primary endpoint

Domain 3: Sample size based on a clinically relevant effect

Labels	Decision rules
Yes	use of an absolute minimal important difference (MID) or a clinically meaningful difference (i.e., Cohen's <i>d</i>) for the primary endpoint(s) in any resource. If >1 resource, concordance is required.
No	- use of a difference based on statistically significant results from previous studies - use of a difference without justification

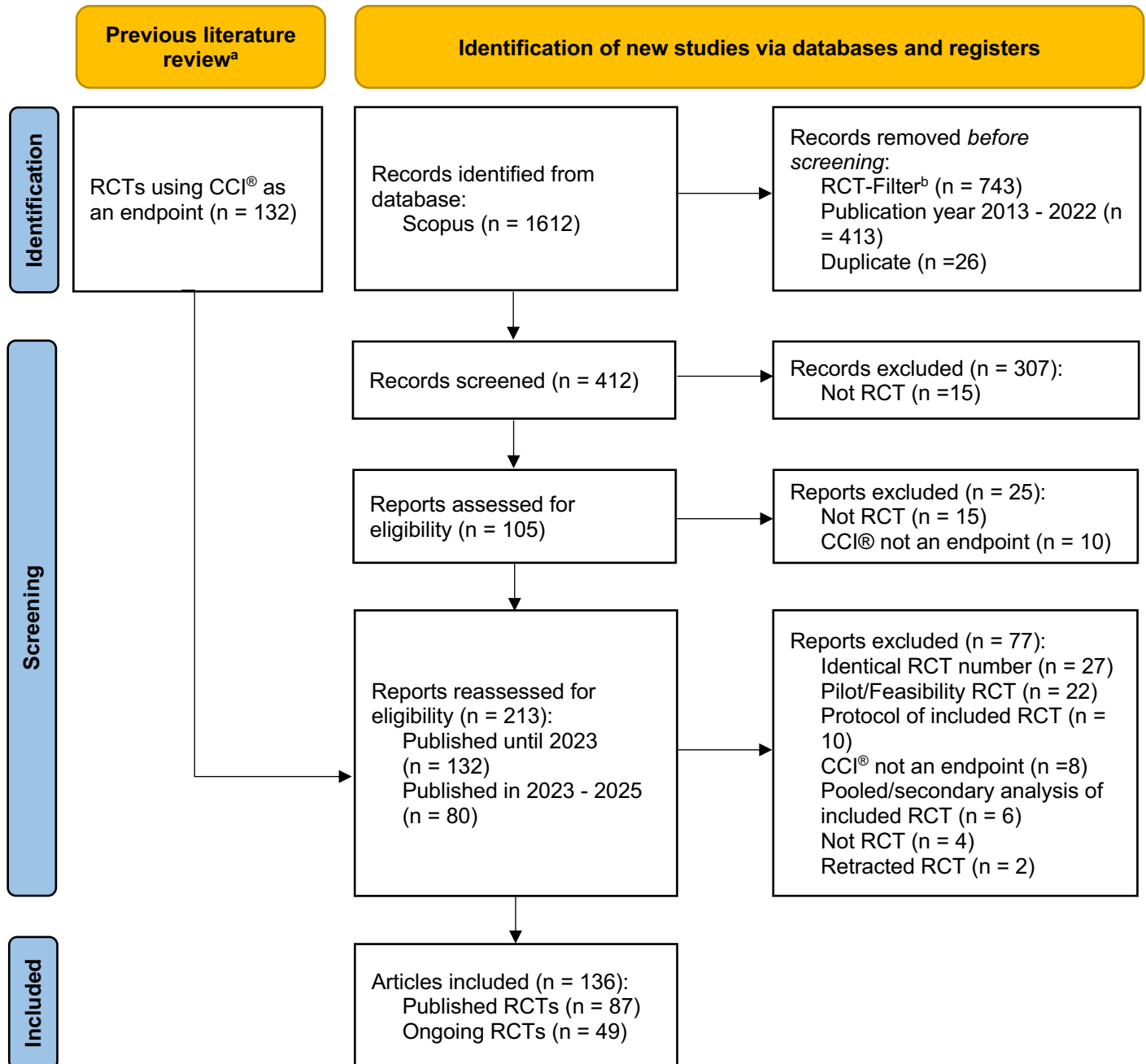
Discordant between pretrial document and final RCT	• mismatch in the nature of the difference used (meaningful/ statistical)
Not applicable	• no report of specific primary endpoint(s) (condition 1 not fulfilled) • no report of sample size calculation • incomplete report of sample size calculation, i.e., no report of difference used

Domain 4: Discussion of the clinical relevance of the results

Labels	Decision rules
Yes	• discussion of the results for the primary outcome(s) in terms of clinical relevance • discussion of the observed results for the primary outcome(s) in light with the MID, either from the literature, or from the predefined MID/ Cohen's d for the sample size calculation if criterium 3 fulfilled
No	• discussion of the results for the primary outcome(s) in terms of statistical relevance
Not applicable	• no report of specific primary endpoint(s) (condition 1 not fulfilled) • description of trial design and/or statistical analyses not aligned with a specific hypothesis for primary endpoint(s) in any resource

Online-Only Supplement 3 – eFigure 1

eFigure 1 PRIMA Flow chart for study inclusion for pilot phase.



^a Systematic literature review conducted on August 16, 2023. ¹

^b RCT Filter from National University of Singapore (NUS) medical Library. ²

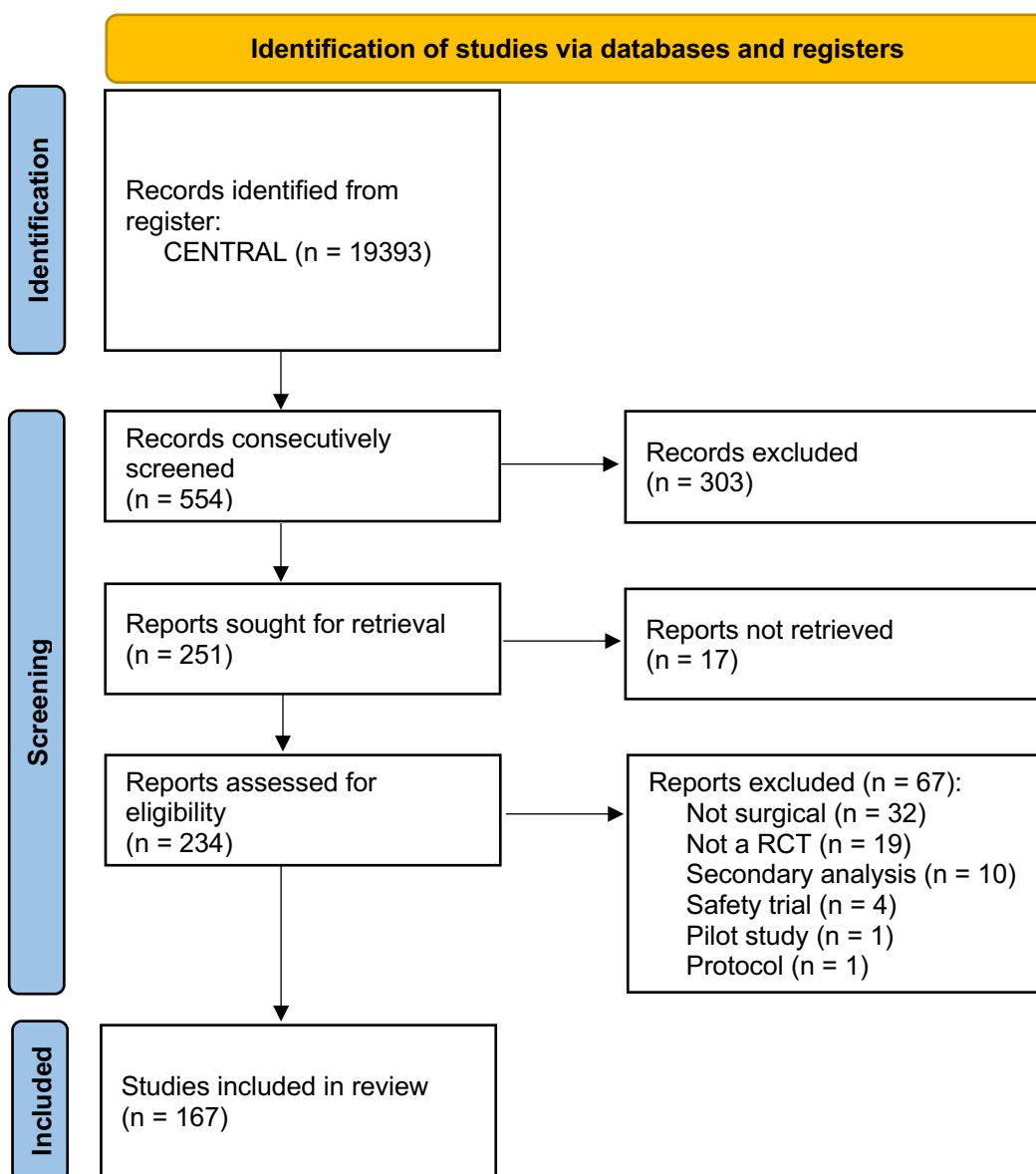
Source: Page MJ, et al. BMJ 2021;372:n71. doi: 10.1136/bmj.n71.

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Abbreviations: CCI®, Comprehensive Complication Index®; RCT, Randomized controlled trial

Online-Only Supplement 4 – eFigure 2

eFigure 2 PRIMA Flow chart for study inclusion for main study.



Source: Page MJ, et al. BMJ 2021;372:n71. doi: 10.1136/bmj.n71.

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Abbreviations: RCT, Randomized controlled trial

Online-Only Supplement 5 – eTable 1

eTable 1 Additional characteristics of surgical randomized controlled trials.

Characteristics	Pilot N = 87^a	Final N = 167^a
Trial hypothesis corresponding to the trial design		
Superiority	77 (89%)	157 (94.0%)
Non-inferiority	8 (9%)	8 (4.8%)
Equivalence	2 (2%)	0 (0.0%)
NA	0 (0%)	2 (1.2%)
Statistical significance achieved for primary outcome(s)		
Yes	41 (47%)	52 (31.1%)
No	42 (48%)	43 (25.7%)
NA	4 (5%)	72 (43.1%)

^a Results are expressed in numbers (percentage(%)).

Abbreviations: NA, Not applicable

References

1. Abbassi F, Pfister M, Lucas KL, et al. Milestones in Surgical Complication Reporting: Clavien-Dindo Classification 20 Years and Comprehensive Complication Index 10 Years. *Ann Surg.* 2024;280(5):763-771. doi:10.1097/SLA.0000000000006471
2. Chin A. LibGuides: Systematic Reviews: Search Filters. Accessed June 8, 2026. <https://libguides.nus.edu.sg/sysreviews/filters>